

Two-grid method for two-dimensional nonlinear Schrödinger equation by finite element method

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A conservative two-grid finite element scheme is presented for the two-dimensional nonlinear Schrödinger equation. One Newton iteration is applied on the fine grid to linearize the fully discrete problem using the coarse-grid solution as the initial guess. Moreover, error estimates are conducted for the two-grid method. It is shown that the coarse space can be extremely coarse, with no loss in the order of accuracy, and still achieve the asymptotically optimal approximation as long as the mesh sizes satisfy $H = O(h^{\frac{1}{2}})$ in the two-grid method. The numerical results show that this method is very effective.

KEYWORDS

conservative, convergence, finite element method, Schrödinger equation, two-grid method

1 | INTRODUCTION

The nonlinear Schrödinger (NLS) equation is one of the most important equations of mathematical physics, with applications in many fields, such as plasma physics, nonlinear optics, water waves, bimolecular dynamics, and protein chemistry. In this article, we consider the following initial-boundary value problem for the two-dimensional NLS equation:

$$i \frac{\partial u}{\partial t} - \Delta u + |u|^2 u = 0, \quad (x, y, t) \in \Omega \times (0, T], \quad (1.1)$$

$$u(x, y, 0) = u_0(x, y), \quad (x, y) \in \overline{\Omega}, \quad (1.2)$$

$$u(x, y, t) = 0, \quad (x, y) \in \Gamma, \quad 0 \leq t \leq T, \quad (1.3)$$

where $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the Laplacian operator, Ω is $[x_l, x_r] \times [y_l, y_r]$, Γ is the boundary of Ω , $u(x, y, t)$ is a complex function, $u_0(x, y)$ is a prescribed smooth complex function, and $i^2 = -1$.

The two conservative laws are obtained by computing the inner product of equation (1.1) with u and $\frac{\partial u}{\partial t}$ and then taking the imaginary part and the real part, as follows

$$Q(t) = \|u(\cdot, t)\| = \|u_0\| = Q(0), \quad (1.4)$$

$$E(t) = \|u(\cdot, t)\|_{H^1}^2 + \frac{1}{2} \|u(\cdot, t)\|_{L^4}^4 = E(0). \quad (1.5)$$

Zhang et al. found that nonconservative schemes are subject to nonlinear blow-up when studying the NLS equation, so they presented a conservative difference scheme in [1]. Furthermore, extensive mathematical and numerical studies have been conducted for the NLS equation [2–9].

In recent years, there has been tremendous interest in developing finite difference schemes and finite element methods for two-dimensional Schrödinger equations [3, 10–16]. Wang studied the split-step finite difference method for various cases of NLS equations, such as cubic NLS equations, coupled NLS equations with constant coefficients and Gross-Pitaevskii equations in 1D, 2D, and 3D in [17]. Dehghan presented a compact split-step finite difference method for solving NLS equations with constant and variable coefficients in [18]. Ref. [10, 13] introduced the alternating direction implicit method for two-dimensional NLS equations, which has the advantage of not increasing the dimensions of the coefficient matrices corresponding to the matrix equations. The multigrid method is an efficient algorithm to solve the NLS equation [19–21].

In this article, we consider an effective algorithm for this essential system. Xu [22, 23] introduced the two-grid finite element approach to efficiently solve nonlinear elliptic equations, where the basic idea is to use a coarse space to produce a rough approximation of the solution and then to use the rough approximately as the initial guess for one Newton iteration on the fine grid. This procedure involves a nonlinear solver on the coarse space and a linear solver on the fine space. It is a simple but effective algorithm that has been applied to many types of problems. Dawson [24] and Chen [25, 26] proposed a two-grid method for quasilinear reaction diffusion equations. Jin et al. [11] successfully extended the two-grid finite element method to solve coupled partial differential equations, such as the linear Schrödinger equation, where the equations on the fine grid are decoupled so that the computational complexity of solving the Schrödinger equation is comparable to solving two decoupled Poisson equations on the same fine grid. Chien et al. [27] proposed two-grid discretization schemes with two-loop continuation algorithms for nonlinear Schrödinger equations, where the centered difference approximations, the six-node triangular elements and the Adini elements are used to discretize the partial differential equations. Numerical experiments have shown that these schemes are efficient, but no rigorous error analysis was presented. Wu [28, 29] constructed two-grid mixed finite element schemes for nonlinear Schrödinger equations, where a linear and indefinite (the typical nature of mixed finite elements) discretization systems are solved on the fine grid. Numerical experiments using these schemes have been shown to be efficient; however, no rigorous error analysis has been conducted. Zhang et al. [30] recently extended the approach given in [11] to time-dependent linear Schrödinger equations, where the error convergence rate is of the order of $O(\Delta t + h + H^2 + \frac{H^2}{\Delta t})$ in the H^1 -norm. Therefore, it is natural to develop a two-grid scheme for time-dependent nonlinear Schrödinger equations (1.1)–1.3 that produces a better error analysis result. We first estimate the finite element solution in the sense of the $L^\infty(J; L^p)$ norm for the Schrödinger equation. Then, we present our main algorithm, a new two-grid method based on one Newton iteration on the fine grid. There are several studies on the NLS equation using different treatments, but, to the best of our knowledge, there are few results on two-grid algorithms used for such nonlinear coupled problems. The main goal of our algorithm is to solve a nonlinear coupled system in the very coarse grid and then solve the decoupled linear system on the fine grid. It is shown that the coarse grid can achieve the asymptotically optimal approximation as long as the mesh sizes satisfy $H = O(h^{\frac{1}{2}})$ in the two-grid algorithm. Under certain

assumptions, the two-grid algorithm can retain the same order of approximation accuracy as the coupled and nonlinear algorithms and with a much lower time cost.

The remainder of this article is organized as follows. We present the weak formulas of our model in Section 2. In Section 3, a simple analysis of the finite element solutions of the model is given. Our main algorithm and its convergence analysis are presented in Section 4. In the final Section, an iterative algorithm and numerical results are discussed.

2 | WEAK FORMULATION OF THE PROBLEM AND PRELIMINARIES

We now present the standard notation that will be used in this article. We denote $W^{m,p}(\Omega)$ the complex Sobolev space on Ω with norm $\|\cdot\|_{m,p}$ given by $\|\phi\|_{m,p} = \sum_{|\alpha| \leq m} \|D^\alpha \phi\|_{L^p}^p$. We set $H^m(\Omega) = W^{m,2}(\Omega)$, $\|\cdot\|_m = \|\cdot\|_{m,2}$, $\|\cdot\|_\infty = \|\cdot\|_{L^\infty}$ and $\|\cdot\| = \|\cdot\|_{0,2}$, and the subspace of $H^1(\Omega)$ consisting of functions with vanishing trace on $\partial\Omega$ by $H_0^1(\Omega)$.

Let $(\cdot, \cdot)$ be the inner product on $L^2(\Omega) = H^0(\Omega)$ defined by

$$(u, v) = \int_{\Omega} u(x) \bar{v}(x) dx, \quad \text{for } u, v \in L^2(\Omega),$$

and denote by $\|\cdot\|$ the associated L^2 norm.

For any complex-valued function $v = v_1 + iv_2$, let

$$\begin{aligned} \|v\|_{m,p} &= (\|v_1\|_{m,p}^p + \|v_2\|_{m,p}^p)^{\frac{1}{p}}, \\ \|v\|_{m,\infty} &= \|v_1\|_{m,\infty} + \|v_2\|_{m,\infty}. \end{aligned}$$

The weak solution of problem (1.1)–(1.3) is defined as follows: Find $u \in H_0^1$ such that

$$i \left(\frac{\partial u}{\partial t}, \chi \right) + (\nabla u, \nabla \chi) + (|u|^2 u, \chi) = 0, \quad \text{for } \forall \chi \in H_0^1. \quad (2.1)$$

Let $\Delta t = T/N$ and $t^n = n\Delta t$. Define $\phi^n = \phi(\cdot, n)$, $\partial_t \phi^n = \frac{\phi^n - \phi^{n-1}}{\Delta t}$, and discrete L^2 -norm

$$\begin{aligned} \|\phi\|_{l^2(0,T)} &= (\sum_{n=1}^N \|\phi^n\|^2 \Delta t)^{\frac{1}{2}}, \\ \|\phi\|_{l^2(0,t^m)} &= (\sum_{n=1}^m \|\phi^n\|^2 \Delta t)^{\frac{1}{2}}, \end{aligned}$$

where $0 < m < N$.

The full discrete procedure of (1.1)–(1.3) at $t = t^n$ is given by finding u_h^n such that $u_h(x, 0) = u_h^0$.

$$i(\partial_t u_h^n, \chi) + (\nabla u_h^{n-\frac{1}{2}}, \nabla \chi) + (|u_h^{n-\frac{1}{2}}|^2 u_h^{n-\frac{1}{2}}, \chi) = 0, \quad \text{for } \forall \chi \in H_0^1, \quad (2.2)$$

where $u_h^{n-\frac{1}{2}} = \frac{u_h^{n-1} + u_h^n}{2}$.

In this article, we assume that u , the solution of (2.1), is sufficiently smooth. Let $\mathcal{T}_h$ be a family of regularity finite element partitions of Ω . Let the corresponding finite element space $M_h \subset H_0^1(\Omega)$, and throughout this article, C denotes a positive constant independent of h but may have different values

at different locations. We assume that M_h satisfies the following approximation property: there exists an integer $l \geq 1$ such that

$$\inf_{z_h \in M_h} (\|z - z_h\|_{L^2(\Omega)} + h\|z - z_h\|_{H^1(\Omega)}) \leq Ch^{l+1}\|z\|_{H^{l+1}(\Omega)}, \quad \text{for } z \in H^{l+1}(\Omega) \cap H_0^1(\Omega). \quad (2.3)$$

Now, we introduce an existence and uniqueness lemma from [2].

Lemma 2.1 *Suppose that $\|u_h^0\|^2 \leq \frac{1}{4}$; then, there exists a unique solution u_h^n of the scheme (2.2) that satisfies the two conservative laws*

$$Q^n = \|u_h^n\| = \|u_h^{n-1}\| = \cdots = \|u_h^0\|. \quad (2.4)$$

$$E^n = \|u_h^n\|_{H^1}^2 + \frac{1}{2}\|u_h^n\|_{L^4}^4 = \cdots = \|u_h^0\|_{H^1}^2 + \frac{1}{2}\|u_h^0\|_{L^4}^4. \quad (2.5)$$

Lemma 2.2 (Grönwall's inequality [31]) *Suppose that the discrete mesh function $\{w_n, n = 1, 2, \dots, N\}$, $N\Delta t = T$, satisfies the inequality*

$$w_n \leq A + \sum_{k=1}^n B_k w_k \Delta t, \quad 0 \leq n\Delta t \leq T,$$

where A and B_k ($k = 1, 2, \dots, N$) are non-negative constants. Then,

$$\|w_n\|_\infty \leq Ae^{2\sum_{k=1}^N B_k \Delta t}, \quad (2.6)$$

where Δt is sufficiently small, such that $\Delta t(\max_{k=1,2,\dots,N} B_k) \leq \frac{1}{2}$.

3 | L^P ERROR ESTIMATE OF THE FINITE ELEMENT METHOD

In this section, we analyze the convergence of the error estimate for the finite element solution. We define the elliptic projection operator $P_h : H_0^1 \rightarrow M_h$ by

$$(\nabla(u - P_h u), \nabla \chi) + \lambda(u - P_h u, \chi) = 0, \quad \chi \in M_h. \quad (3.1)$$

First, we achieve a lemma with the help of an elliptic problem.

Lemma 3.1 *$P_h u^n$ is the elliptic projection at $t = t^n$ with $1 \leq n \leq N$ defined in (3.1); then, for $2 \leq q < \infty$, we have*

$$\|u^n - P_h u^n\|_{L^p} \leq C\|u\|_{W^{l+1,p}} h^{l+1}. \quad (3.2)$$

Proof Define elliptic operator L as

$$Lu^n = -\nabla \cdot (\nabla u^n) + \lambda u^n, \quad (3.3)$$

and the corresponding bilinear form as

$$A(u^n, \chi) = (\nabla u^n, \nabla \chi) + (\lambda u^n, \chi), \quad (3.4)$$

where λ is chosen to guarantee the coercivity of the bilinear form $A(u, \chi)$ such that

$$A(u^n, u^n) \geq \gamma \|u^n\|_1^2,$$

where γ is a positive constant. From the formulation (3.1), we know that $P_h u^n$ is in fact the finite element solution of the elliptic problem. To prove (3.2), we use a duality argument by considering the auxiliary problem

$$L\omega = \text{sgn}(u^n - P_h u^n) |u^n - P_h u^n|^{p-1}, \text{ in } \Omega, \quad (3.5)$$

$$\omega = 0, \text{ on } \partial\Omega. \quad (3.6)$$

is uniquely solvable for $\omega \in L^p$ and has the following regularity

$$\|\omega\|_{2,q} \leq C \|L\omega\|_q = C \|u^n - P_h u^n\|_p^{p-1}. \quad (3.7)$$

Using (3.1), Hölder's inequality, (2.3) and (3.7)

$$\begin{aligned} \|u^n - P_h u^n\|_{L^p}^p &= A(u^n - P_h u^n, \omega) \\ &= A(u^n - P_h u^n, \omega - I_h \omega) \\ &\leq C \|u^n - P_h u^n\|_{1,p} \|\omega - I_h \omega\|_{1,q} \\ &\leq Ch \|u^n - P_h u^n\|_{1,p} \|\omega\|_{2,q} \\ &\leq Ch \|u^n - P_h u^n\|_{1,p} \|u^n - P_h u^n\|_p^{p-1}, \end{aligned}$$

where I_h is an interpolation operator.

We obtain

$$\|u^n - P_h u^n\|_{1,p} \leq \inf_{u_h \in M_h} \|u^n - u_h^n\|_{1,p} \leq Ch^l \|u\|_{W^{l+1,p}}$$

from Theorem 8.53 of [32] and the approximation property; then, we have

$$\|u^n - P_h u^n\|_{L^p} \leq Ch \|u^n - P_h u^n\|_{1,p} \leq Ch^{l+1} \|u\|_{W^{l+1,p}}.$$

Differentiation of (3.1) with respect to time leads to the additional estimate

$$\|\partial_t(u - P_h u)\|_{L^p} \leq M \left(\|u\|_{l+1} + \left\| \frac{\partial u}{\partial t} \right\|_{H^{l+1}} \right) h^{l+1}. \quad (3.8)$$

We now obtain the L^2 error estimate for $\|P_h u^n - u_h^n\|$. ■

Lemma 3.2 $P_h u^n$ is the elliptic projection at $t = t^n$ with $1 \leq n \leq N$ defined in (3.1), and u_h^n is the Galerkin solution that satisfies the equation (2.2). Then, for $2 \leq q < \infty$, we have

$$\|P_h u^n - u_h^n\| \leq C(h^{l+1} + (\Delta t)^2). \quad (3.9)$$

Proof By adding (3.1), (2.1), and subtracting (2.2) at $t = t^{n-\frac{1}{2}}$, we obtain

$$\begin{aligned} & i \left(\frac{\partial u^{n-\frac{1}{2}}}{\partial t} - \partial_t u_h^n, \chi \right) + (\nabla(P_h u^{n-\frac{1}{2}} - u_h^{n-\frac{1}{2}}), \nabla \chi) \\ &= -\lambda(P_h u^{n-\frac{1}{2}} - u^{n-\frac{1}{2}}, \chi) - \frac{1}{8}(|u^{n-1} + u^n|^2(u^n + u^{n-1}) \\ & \quad - |u_h^n + u_h^{n-1}|^2(u_h^n + u_h^{n-1}), \chi). \end{aligned} \quad (3.10)$$

We then estimate (3.10). Let $\rho^n = P_h u^n - u_h^n$, $\eta = u - P_h u$ and $v = \rho^{n-\frac{1}{2}}$; then, we can rewrite (3.10) as

$$\begin{aligned} i(\partial_t \rho^n, \rho^{n-\frac{1}{2}}) + (\nabla \rho^{n-\frac{1}{2}}, \nabla \rho^{n-\frac{1}{2}}) &= i \left(\frac{\partial u^{n-\frac{1}{2}}}{\partial t} - \partial_t u^n, \rho^{n-\frac{1}{2}} \right) - i(\partial_t \eta^n, \rho^{n-\frac{1}{2}}) \\ & \quad - \lambda(\eta^{n-\frac{1}{2}}, \rho^{n-\frac{1}{2}}) - (f(u^{n-\frac{1}{2}}) - f(u_h^{n-\frac{1}{2}}), \rho^{n-\frac{1}{2}}), \end{aligned} \quad (3.11)$$

where $f(u) = |u|^2 u$.

Taking the imaginary part, we obtain

$$\frac{\|\rho^n\|^2 - \|\rho^{n-1}\|^2}{2\Delta t} = (F_1 + F_2 + F_3 + F_4, \rho^{n-\frac{1}{2}}). \quad (3.12)$$

Each term on the right side of (3.12) is bounded as

$$|(F_1, \rho^{n-\frac{1}{2}})| = \operatorname{Re} \left(\frac{\partial u^{n-\frac{1}{2}}}{\partial t} - \partial_t u^n, \rho^{n-\frac{1}{2}} \right) \leq C \left\| \frac{\partial^3 u}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 (\Delta t)^3 + \|\rho^{n-\frac{1}{2}}\|^2, \quad (3.13)$$

$$\begin{aligned} |(F_2, \rho^{n-\frac{1}{2}})| &= \operatorname{Re} |(\partial_t \eta^n, \rho^{n-\frac{1}{2}})| \leq C \frac{1}{\Delta t} \int_{t^{n-1}}^{t^n} \left\| \frac{\partial \eta}{\partial t} \right\| dt \cdot \|\rho^{n-\frac{1}{2}}\| \\ &\leq C \frac{1}{\Delta t} \left\| \frac{\partial \eta}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 + \|\rho^{n-\frac{1}{2}}\|^2, \end{aligned} \quad (3.14)$$

$$|(F_3, \rho^{n-\frac{1}{2}})| = \operatorname{Im} |\lambda(\eta^{n-\frac{1}{2}}, \rho^{n-\frac{1}{2}})| \leq \|\eta^{n-\frac{1}{2}}\|^2 + \|\rho^{n-\frac{1}{2}}\|^2 \leq Ch^{2l+2} + \|\rho^{n-\frac{1}{2}}\|^2, \quad (3.15)$$

$$\begin{aligned} |(F_4, \rho^{n-\frac{1}{2}})| &= \operatorname{Im} \left| -\frac{1}{8}(|u^{n-1} + u^n|^2(u^n + u^{n-1}) - |u_h^n + u_h^{n-1}|^2(u_h^n + u_h^{n-1}), \rho^{n-\frac{1}{2}}) \right| \\ &\leq (|u^{n-\frac{1}{2}}|^2(\rho^{n-\frac{1}{2}} + \eta^{n-\frac{1}{2}}) + [|u^{n-\frac{1}{2}}|^2 - |u_h^{n-\frac{1}{2}}|^2]u_h^{n-\frac{1}{2}}, \rho^{n-\frac{1}{2}}) \\ &\leq C((\rho^{n-\frac{1}{2}} + \eta^{n-\frac{1}{2}}) + [| \rho^{n-\frac{1}{2}} | + | \eta^{n-\frac{1}{2}} |], \rho^{n-\frac{1}{2}}) \\ &\leq C(\|\rho^{n-\frac{1}{2}}\|^2 + h^{2l+2}), \end{aligned} \quad (3.16)$$

where we make the induction hypothesis that

$$\|u_h^n\|_\infty \leq C. \quad (3.17)$$

Then, from (3.12) to (3.16), we have

$$\begin{aligned} \frac{\|\rho^n\|^2 - \|\rho^{n-1}\|^2}{2\Delta t} &\leq C \left[\left\| \frac{\partial^3 u}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 (\Delta t)^3 + \frac{1}{\Delta t} \left\| \frac{\partial \eta}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 \right. \\ &\quad \left. + \|\rho^n\|^2 + \|\rho^{n-1}\|^2 + h^{2l+2} \right]. \end{aligned} \quad (3.18)$$

By multiplying $2\Delta t$ on both sides of the last inequality, summing from $n = 1$ through $n = m$ and noting that $R_h u_0 = u_0$, we obtain

$$\|\rho^m\|^2 \leq C[h^{2l+2} + (\Delta t)^4] + C\Delta t \sum_{n=1}^m \|\rho^n\|^2. \quad (3.19)$$

According to Lemma 2.2, when Δt is sufficiently small

$$\|\rho^n\|^2 \leq C[h^{2l+2} + (\Delta t)^4]. \quad (3.20)$$

Note that (3.2) and (3.20) imply that the induction hypothesis (3.17) holds for small h , so the entire argument is valid. $\blacksquare$

4 | TWO-GRID ALGORITHM AND ERROR ESTIMATE

In this section, we present the two-grid algorithm and analyze the convergence accuracy of the scheme. The fundamental ingredient in this scheme is another finite element space $M_H (\subset M_h)$ ($h < H < 1$) defined on a coarser quasiuniform triangulation or rectangulation of Ω . For $n \geq 1$, let in (2.2) $U_h^n = V_h^n + iW_h^n$, V_h^n , and $W_h^n \in \text{RM}_h$, where RM_h denotes the space of the real-valued elements of M_h . Now, we present the two-grid algorithm based on one implicit Newton iteration on the fine grid. The two-grid algorithm consists of two main steps.

Step 1. On the coarse grid $\mathcal{T}_H$, solve the following nonlinear coupling system for $v_H^n, w_H^n \in \text{RM}_H$

$$(\partial_t v_H^n, \chi) = -(\nabla w_H^{n-\frac{1}{2}}, \nabla \chi) - ([v_H^{n-\frac{1}{2}}]^2 + [w_H^{n-\frac{1}{2}}]^2) w_H^{n-\frac{1}{2}}, \chi, \quad \text{for } \forall \chi \in \text{RM}_H, \quad (4.1)$$

$$(\partial_t w_H^n, \chi) = (\nabla v_H^{n-\frac{1}{2}}, \nabla \chi) + ([v_H^{n-\frac{1}{2}}]^2 + [w_H^{n-\frac{1}{2}}]^2) v_H^{n-\frac{1}{2}}, \chi, \quad \text{for } \forall \chi \in \text{RM}_H, \quad (4.2)$$

Step 2. On the fine grid $\mathcal{T}_h$, compute $V_h^n, W_h^n \in \text{RM}_h$ to satisfy the following linear decoupling system

$$\begin{aligned} (\partial_t V_h^n, \varphi) &= -\frac{1}{2}(\nabla(W_h^n + W_h^{n-1}), \nabla \varphi) \\ &\quad - \frac{1}{4}((v_H^n + v_H^{n-1})(w_H^n + W_h^{n-1})(V_h^n - v_H^n), \varphi) \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{8}([(v_H^n + V_h^{n-1})^2 + 3(w_H^n + W_h^{n-1})^2](W_h^n - w_H^n), \varphi) \\
& -\frac{1}{8}([(v_H^n + V_h^{n-1})^2 + (w_H^n + W_h^{n-1})^2](w_H^n + W_h^{n-1}), \varphi), \\
& \text{for } \forall \varphi \in RM_h, \tag{4.3} \\
(\partial_t W_h^n, \varphi) &= \frac{1}{2}(\nabla(V_h^n + V_h^{n-1}), \nabla \varphi) \\
& + \frac{1}{4}((w_H^n + W_h^{n-1})(v_H^n + V_h^{n-1})(W_h^n - w_H^n), \varphi) \\
& + \frac{1}{8}([(w_H^n + W_h^{n-1})^2 + 3(v_H^n + V_h^{n-1})^2](V_h^n - v_H^n), \varphi) \\
& + \frac{1}{8}([(w_H^n + W_h^{n-1})^2 + (v_H^n + V_h^{n-1})^2](v_H^n + V_h^{n-1}), \varphi), \\
& \text{for } \forall \varphi \in RM_h. \tag{4.4}
\end{aligned}$$

Remark In the two-grid algorithm, this article uses the coarse grid to produce a rough approximation of solutions for the nonlinear coupling equations separated into the real and imaginary parts and then uses the rough approximations as the initial guesses for one Newton iteration on the fine grid.

For simplification, we define L such that for $g : R^2 \rightarrow R$, $g(x, y) = (x^2 + y^2)y$, there holds

$$\max|\partial_1 g(x, y)| + \max|\partial_2 g(x, y)| \leq L,$$

where we denote $\partial_1 g = g_x$, $\partial_2 g = g_y$.

By combining Lemma 3.1 and 3.2, we obtain the following result.

Lemma 4.1 Suppose that u^n is the exact solution satisfying (2.1) at $t = t^n$, and $u_H^n = v_H^n + iw_H^n$ is the two-grid solution satisfying equation (4.1)–(4.2). If we set $R_H v_0 = v_H^0, R_H w_0 = w_H^0$, then for $1 \leq n \leq N$, we have

$$\|u^n - u_H^n\| \leq C((\Delta t)^2 + H^{l+1}). \tag{4.5}$$

Now, we analyze the error estimation of the two-grid solution of step 2. In the following analysis, we need to prove $\|u^n - U_h^n\|$.

Theorem 4.1 Suppose that u^n is the exact solution that satisfies (2.1) at $t = t^n$, and $U_h^n = V_h^n + iW_h^n$ is the two-grid solution that satisfies equations (4.3)–(4.4). If we select $R_h u_0 = U_h^0$, then for $1 \leq n \leq N$, we have

$$\|u^n - U_h^n\| \leq C(h^{l+1} + H^{2l+2} + (\Delta t)^2). \tag{4.6}$$

Proof We use the elliptic projection as an intermediate variable to assist the proof. Add (2.1), (3.1); then, the equation can be rewritten as a system of two equations for $v^{n-\frac{1}{2}}$

and $w^{n-\frac{1}{2}}$. Then subtract (4.3) from the first of the two resulting equations to obtain the following relationship

$$\begin{aligned} \left(\frac{\partial v^{n-\frac{1}{2}}}{\partial t} - \partial_t V_h^n \right), \varphi &= -(\nabla(P_h w^{n-\frac{1}{2}} - W_h^{n-\frac{1}{2}}), \nabla \varphi) - \lambda(P_h w^{n-\frac{1}{2}} - w^{n-\frac{1}{2}}, \varphi) \\ &\quad - (g(v^{n-\frac{1}{2}}, w^{n-\frac{1}{2}}) + g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})) \\ &\quad + \frac{1}{2} \partial_1 g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})(V_h^n - v_H^n), \varphi \\ &\quad + \frac{1}{2} \partial_2 g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})(W_h^n - w_H^n), \varphi, \end{aligned} \quad (4.7)$$

where $\bar{v}_H^{n-\frac{1}{2}} = \frac{v_H^n + V_h^{n-1}}{2}$, $\bar{w}_H^{n-\frac{1}{2}} = \frac{w_H^n + W_h^{n-1}}{2}$.

Let $\xi^n = P_h v^n - V_h^n$, $\zeta^n = P_h w^n - W_h^n$, $\eta_1^{n-\frac{1}{2}} = P_h v^{n-\frac{1}{2}} - v^{n-\frac{1}{2}}$, $\eta_2^{n-\frac{1}{2}} = P_h w^{n-\frac{1}{2}} - w^{n-\frac{1}{2}}$; then, we have

$$\begin{aligned} &\left(\frac{\xi^n - \xi^{n-1}}{\Delta t}, \varphi \right) + (\nabla \zeta^{n-\frac{1}{2}}, \nabla \varphi) \\ &= - \left(\frac{\partial v^{n-\frac{1}{2}}}{\partial t} - \frac{v^n - v^{n-1}}{\Delta t}, \varphi \right) - \left(\frac{\eta_1^n - \eta_1^{n-1}}{\Delta t}, \varphi \right) + \lambda(\eta_2^{n-\frac{1}{2}}, \varphi) \\ &\quad - \frac{1}{2} (\partial_1 g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})(v^n - V_h^n) + \partial_2 g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})(w^n - W_h^n), \varphi) \\ &\quad + \frac{1}{8} (\partial_1^2 g(v^*, w^*)(v^n - v_H^n)^2 + \partial_2^2 g(v^*, w^*)(w^n - w_H^n)^2 \\ &\quad + 2\partial_1 \partial_2 g(v^*, w^*)(v^n - v_H^n)(w^n - w_H^n), \varphi) \\ &= (Q_1 + Q_2 + Q_3 + Q_4 + Q_5, \varphi), \end{aligned} \quad (4.8)$$

where we use second-order Taylor expansion at point $(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})$ for $g(v^{n-\frac{1}{2}}, w^{n-\frac{1}{2}})$.

Let $\varphi = \xi^{n-\frac{1}{2}}$; then, we have

$$|(Q_1, \xi^{n-\frac{1}{2}})| \leq K \left\| \frac{\partial^3 v^n}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 (\Delta t)^3 + \|\xi^{n-\frac{1}{2}}\|^2, \quad (4.9)$$

$$|(Q_2, \xi^{n-\frac{1}{2}})| \leq \frac{K}{\Delta t} \left\| \frac{\partial \eta_1^n}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 + \|\xi^{n-\frac{1}{2}}\|^2, \quad (4.10)$$

$$|(Q_3, \xi^{n-\frac{1}{2}})| \leq C \|\eta_2^{n-\frac{1}{2}}\|^2 + \|\xi^{n-\frac{1}{2}}\|^2, \quad (4.11)$$

$$\begin{aligned} |(Q_4, \xi^{n-\frac{1}{2}})| &\leq C |(\partial_1 g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})(v^n - V_h^n) + \partial_2 g(\bar{v}_H^{n-\frac{1}{2}}, \bar{w}_H^{n-\frac{1}{2}})(w^n - W_h^n), \xi^{n-\frac{1}{2}})| \\ &\leq L(\|v^n - V_h^n\| + \|w^n - W_h^n\|) \|\xi^{n-\frac{1}{2}}\| \\ &\leq C(\|v^n - P_h v^n + P_h v^n - V_h^n\| + \|w^n - P_h w^n + P_h w^n - W_h^n\|) \cdot \|\xi^{n-\frac{1}{2}}\| \\ &\leq C(h^{2l+2} + (\Delta t)^4 + \|\xi^n\|^2 + \|\zeta^n\|^2 + \|\xi^{n-\frac{1}{2}}\|^2), \end{aligned} \quad (4.12)$$

$$\begin{aligned} |(Q_5, \xi^{n-\frac{1}{2}})| &\leq C |(\partial_1^2 g(v^*, w^*)(v^n - v_H^n)^2 + \partial_2^2 g(v^*, w^*)(w^n - w_H^n)^2 \\ &\quad + 2\partial_1 \partial_2 g(v^*, w^*)(v^n - v_H^n)(w^n - w_H^n), \xi^{n-\frac{1}{2}})| \end{aligned}$$

$$\begin{aligned}
&\leq C(\|v^n - v_H^n\|_4^4 + \|w^n - w_H^n\|_4^4 + \|\xi^{n-\frac{1}{2}}\|^2) \\
&\leq C(\|v^n - P_h v^n + P_h v^n - v_H^n\|_4^4 + \|w^n - P_h w^n + P_h w^n - w_H^n\|_4^4 + \|\xi^{n-\frac{1}{2}}\|^2) \\
&\leq C(h^{2l+2} + H^{-2}H^{4l+4} + (\Delta t)^4 + \|\xi^{n-\frac{1}{2}}\|^2),
\end{aligned} \tag{4.13}$$

where it is proved that the first three inequalities are similar to Lemma 3.2.

Then, from (4.8) to (4.13), we have

$$\begin{aligned}
&\frac{\|\xi^n\|^2 - \|\xi^{n-1}\|}{2\Delta t} + (\nabla \xi^{n-\frac{1}{2}}, \nabla \xi^{n-\frac{1}{2}}) \\
&\leq C \left(\left\| \frac{\partial^3 v^n}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 (\Delta t)^3 + \frac{1}{\Delta t} \left\| \frac{\partial \eta_1}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 \right. \\
&\quad + \|\xi^n\|^2 + \|\xi^{n-1}\|^2 + \|\eta^{n-1}\|^2 + \|\eta^n\|^2 \\
&\quad \left. + \|\zeta^n\|^2 + \|\zeta^{n-1}\|^2 + h^{2l+2} + H^{-2}H^{4l+4} \right).
\end{aligned} \tag{4.14}$$

In an entirely analogous manner, a similar estimate may be obtained for $R_h w^n - W_h^n$.

$$\begin{aligned}
&\frac{\|\zeta^n\|^2 - \|\zeta^{n-1}\|}{2\Delta t} - (\nabla \xi^{n-\frac{1}{2}}, \nabla \zeta^{n-\frac{1}{2}}) \\
&\leq C \left(\left\| \frac{\partial^3 w^n}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 (\Delta t)^3 + \frac{1}{\Delta t} \left\| \frac{\partial \eta_2}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 \right. \\
&\quad + \|\xi^n\|^2 + \|\xi^{n-1}\|^2 + \|\eta^{n-1}\|^2 + \|\eta^n\|^2 \\
&\quad \left. + \|\zeta^n\|^2 + \|\zeta^{n-1}\|^2 + h^{2l+2} + H^{-2}H^{4l+4} \right).
\end{aligned} \tag{4.15}$$

By adding (4.14) and (4.15), we obtain

$$\begin{aligned}
\frac{\|\mu^n\|^2 - \|\mu^{n-1}\|}{2\Delta t} &\leq C \left(\left\| \frac{\partial^3 u^n}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 (\Delta t)^3 + \frac{1}{\Delta t} \left\| \frac{\partial \eta}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 \right. \\
&\quad \left. + \|\mu^n\|^2 + \|\mu^{n-1}\|^2 + h^{2l+2} + H^{-2}H^{4l+4} \right),
\end{aligned} \tag{4.16}$$

where $\|\mu^n\| = \|\xi^n\| + \|\zeta^n\|$.

Multiplying $2\Delta t$ on both sides of this inequality and summing from $n = 1$ to $n = m$ gives

$$\begin{aligned}
\|\mu^m\|^2 &\leq C((\Delta t)^4 \sum_{n=1}^m \left\| \frac{\partial^3 u^n}{\partial t^3} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 + \sum_{n=1}^m \left\| \frac{\partial \eta}{\partial t} \right\|_{L^2(t^{n-1}, t^n; L^2)}^2 \\
&\quad + h^{2l+2} + H^{4l+4}) + \Delta t \sum_{n=1}^m \|\mu^n\|^2 \\
&\leq C(h^{2l+2} + H^{4l+4} + (\Delta t)^4) + \Delta t \sum_{n=1}^m \|\mu^n\|^2.
\end{aligned} \tag{4.17}$$

TABLE 1 Error and CPU time of the Newton iterative method.

H	h	$\ v^n - V_h^n\ $	Rate	$\ w^n - W_h^n\ $	Rate	CPU time
$\frac{1}{2}$	$\frac{1}{4}$	1.4120e-1		1.6520e-1		5.10s
$\frac{1}{4}$	$\frac{1}{16}$	1.2707e-2	2	1.3217e-2	2	35.41s
$\frac{1}{8}$	$\frac{1}{64}$	8.6126e-4	2	8.8858e-4	2	780.20s
$\frac{1}{16}$	$\frac{1}{256}$	5.3639e-5	2	5.4653e-5	2	2556.56s

Then, using the discrete *Grönwall* inequality gives

$$\|\mu^m\|^2 \leq C(h^{2l+2} + H^{4l+4} + (\Delta t)^4). \quad (4.18)$$

We then obtain

$$\|\mu^m\| \leq C(h^{l+1} + H^{2l+2} + (\Delta t)^2). \quad (4.19)$$

From Lemma 3.1 and (4.19), the theorem can be derived using the triangle inequality. $\blacksquare$

Remark We obtain the convergence rate h^{l+1} as long as the mesh sizes satisfy $H = O(h^{\frac{1}{2}})$ in the two-grid algorithm.

5 | NUMERICAL EXPERIMENT

In this section, we construct an example to illustrate the efficiency of the algorithm discussed in Section 4. In this case, we use piecewise linear continuous functions for the numerical solutions.

Example 1 We consider the following example:

$$\begin{aligned}
 i \frac{\partial u}{\partial t} - \Delta u + |u|^2 u &= \left(1 - \frac{2}{x(x-1)} - \frac{2}{y(y-1)} + [xy(x-1)(y-1)]^2\right) u, \\
 (x, y) &\in [-2, 2] \times [-2, 2], \quad t \in [0, 0.01], \\
 u(x, y, t) &= 0, \quad (x, y) \in \Gamma, \quad t \in [0, 0.01], \\
 u(x, y, 0) &= \operatorname{sech}(10x)\operatorname{sech}(10y), \quad (x, y) \in [-2, 2] \times [-2, 2].
 \end{aligned}$$

The exact solution of the equation is

$$u(x, y, t) = \operatorname{sech}(10x)\operatorname{sech}(10y)e^{-it}.$$

For simplicity, the time step $\tau = 1.0e - 4$ and $T = 1.0e - 2$.

TABLE 2 Error and CPU time of the two-grid method.

H	h	$\ v^n - V_h^n\ $	Rate	$\ w^n - W_h^n\ $	Rate	CPU time
$\frac{1}{2}$	$\frac{1}{4}$	1.4110e-1		1.6520e-1		3.48s
$\frac{1}{4}$	$\frac{1}{16}$	1.2698e-2	2	1.3125e-2	2	24.11s
$\frac{1}{8}$	$\frac{1}{64}$	8.5823e-4	2	8.6213e-4	2	483.62s
$\frac{1}{16}$	$\frac{1}{256}$	5.3639e-5	2	5.3653e-5	2	1169.29s

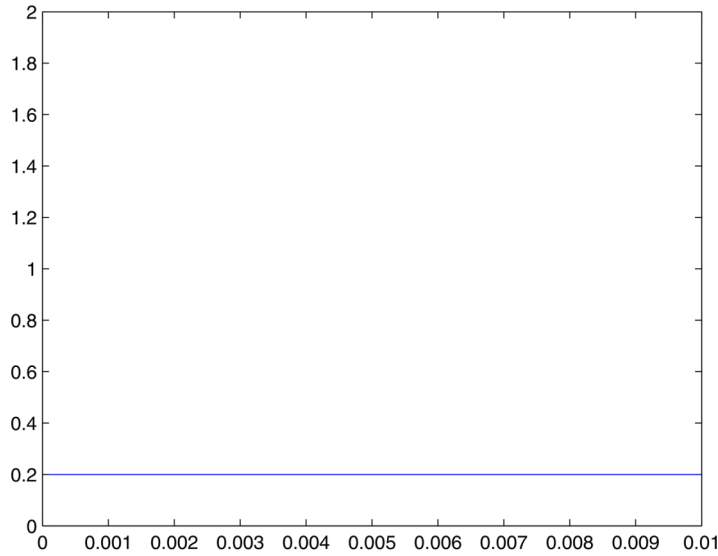


FIGURE 1 Charge conservation of U: $t = 1.0e-2$, $h = 1/64$, $\tau = 1.0e - 4$ [Color figure can be viewed at wileyonlinelibrary.com]

Tables 1 and 2 show the CPU time and errors for $\|v - V_h\|$ and $\|w - W_h\|$. The two-grid method retains the same order of accuracy as the Newton iterative method, but the two-grid method has much lower time cost than the Newton iterative method. As expected, the convergence order of the error for $\|v - V_h\|$ and $\|w - W_h\|$ is second-order.

As shown in Figures 1 and 2, the charge conservation and energy conservation of (2.2) are very good. Figures 3 and 4 shows that the algorithms achieve the asymptotically optimal approximation by applying the two-grid method when the mesh sizes of the coarse grid and fine grid satisfy $H = O(h^{\frac{1}{2}})$. In Figures 5 and 6, we can observe that the error of the two-grid solutions V_h and W_h varies with time.

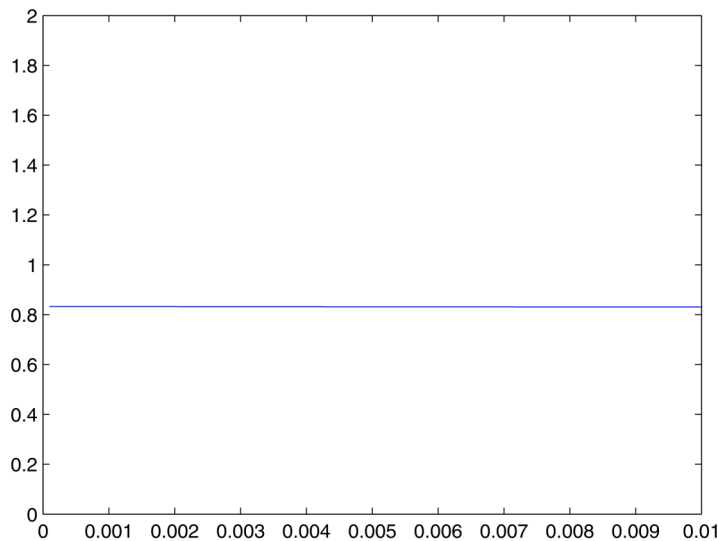


FIGURE 2 Energy conservation of U: $t = 1.0e-2$, $h = 1/64$, $\tau = 1.0e - 4$ [Color figure can be viewed at wileyonlinelibrary.com]

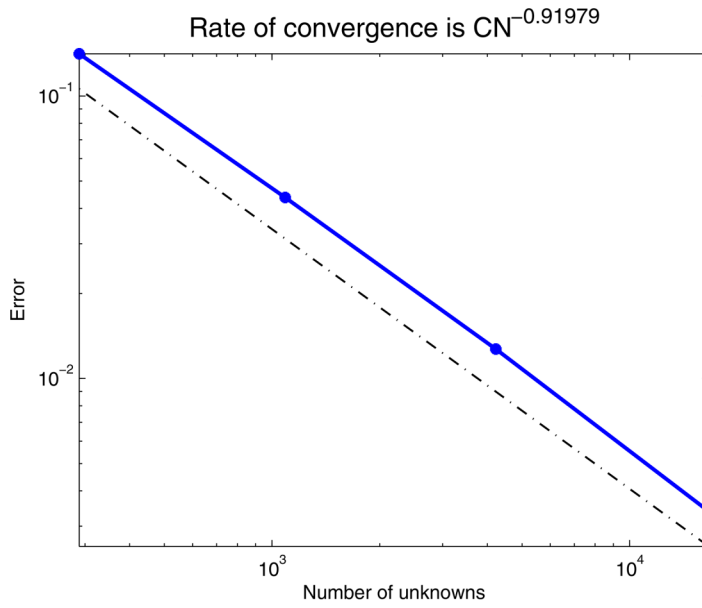


FIGURE 3 Order of two-grid solution v : $t = 1.0e-2$, $\tau = 1.0e - 4$ [Color figure can be viewed at wileyonlinelibrary.com]

6 | CONCLUSION

In this article, a conservative two-grid finite element scheme is presented for two-dimensional nonlinear Schrödinger equations. The proposed method is numerically efficient and enables easy and efficient implementation and software reuse. Numerical analyses and experiments are presented to demonstrate

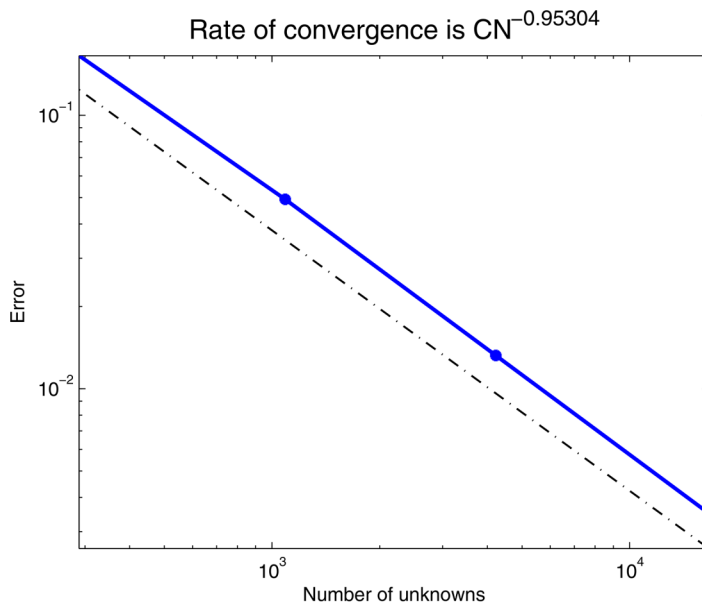


FIGURE 4 Order of two-grid solution w : $t = 1.0e-2$, $\tau = 1.0e - 4$ [Color figure can be viewed at wileyonlinelibrary.com]

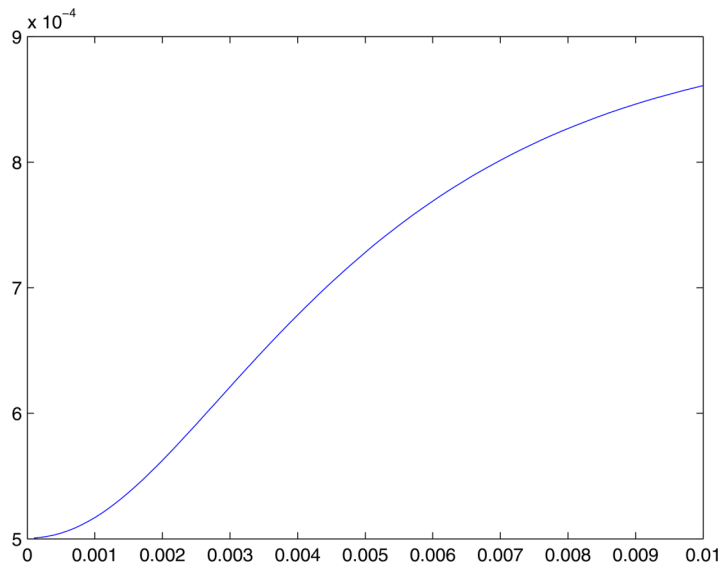


FIGURE 5 Error of v : $t = [0, 0.01]$, $h = 1/64$, $\tau = 1.0e - 4$ [Color figure can be viewed at wileyonlinelibrary.com]

that the two-grid algorithm retains the same order of approximation accuracy as the coupled, nonlinear algorithm. Moreover, it is shown both theoretically and numerically that the coarse space can be extremely coarse and still achieve the asymptotically optimal approximation as long as the mesh sizes satisfy $H = O(h^{\frac{1}{2}})$ in the two-grid algorithm.

It is easy to extend our estimates and algorithm to other derivatives of NLS equations. Future research will be performed to construct and prove the higher accuracy of the scheme and algorithm.

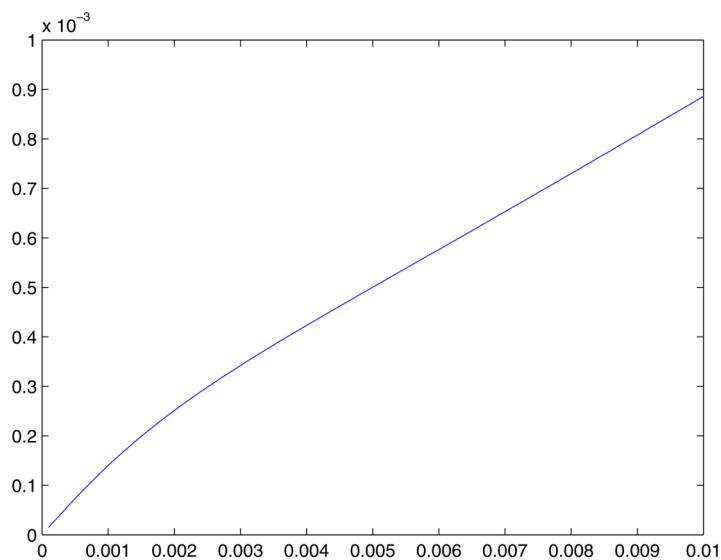


FIGURE 6 Error of w : $t = [0, 0.01]$, $h = 1/64$, $\tau = 1.0e - 4$ [Color figure can be viewed at wileyonlinelibrary.com]

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